

PITCHDECK

Vehicle Selector Pro

A Production-Grade Shopify App for Vehicle Fitment

CLIENT DELIVERABLE · AUGUST 2026

Live demo: <https://vehicle-selector-pro.fly.dev/demo>

Merchant admin: <https://vehicle-selector-pro.fly.dev/demo/admin>

Repository: <https://github.com/ai-dev-2024/vehicle-selector-pro>

Aftermarket parts sell badly online

- A brake pad or wiper blade only fits an exact Year / Make / Model / Trim / Engine (YMMTE) — compatibility is the whole sale.
- Wrong-fit means wrong parts, angry customers, and costly returns. Merchants either bury the data in dense tables or hard-code rules per product.
- Commercial fitment feeds lock merchants into contracts and ongoing per-part fees.
- Verdict: fitment data is kept in silos, rots quickly, and never pays for itself.

THE SOLUTION

One dataset, synced everywhere

- Fitments are entered once into a normalized multi-tenant cache, then pushed to Shopify metafields automatically.
- Customers pick Year → Make → Model → Trim → Engine behind an HMAC-signed App Proxy and only ever see parts that fit.
- Product pages render their own Guaranteed Exact / Does NOT Fit / Universal Fit badge.
- Merchants manage everything from a Polaris admin — no per-part fees, no brittle scripts, no vendor lock-in.

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WHY IT WORKS

A real, browsable demo catalog

48

YMMTE configurations

204

verified fitments

40

mapped products

8

brands represented

Try it live — no setup required. Every number above is visible in the running app.

04

HOW IT WORKS

A cascading fitment engine

- Shopper selects Year → Make → Model → Trim → Engine; each request is HMAC-verified by the App Proxy.
- The local PostgreSQL cache answers instantly — the storefront never waits on Shopify.
- Synced fitments are written to `custom.vehicle_fitment` via GraphQL `metafieldsSet` in batches of 25.
- Product pages, collection filters, and the My Garage feature all stay perfectly in sync.

05

FEATURES

Built to convert and to administer

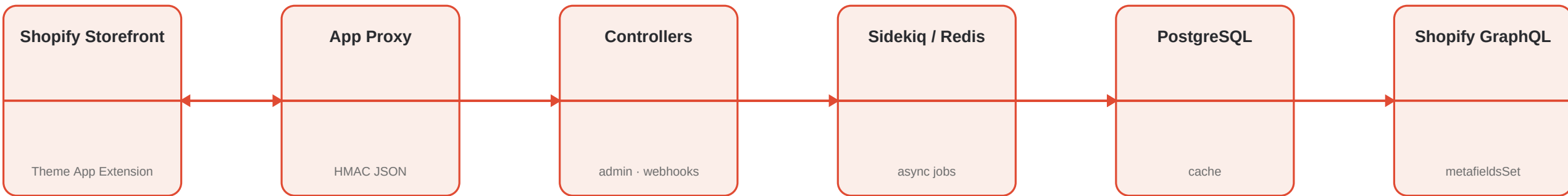
- Exact / No-fit / Universal badges

- My Garage (saved vehicles)

- Mobile-f

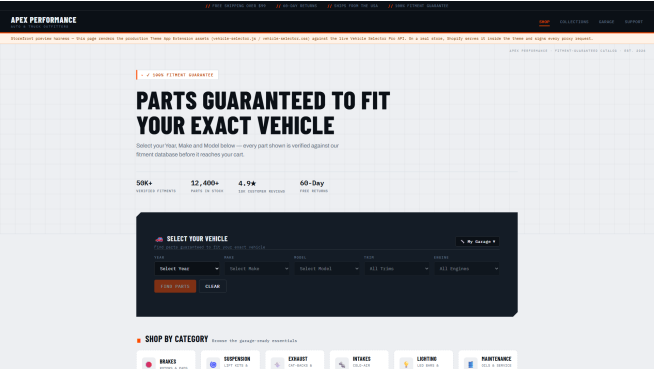
06

Three clean layers, zero ScriptTag

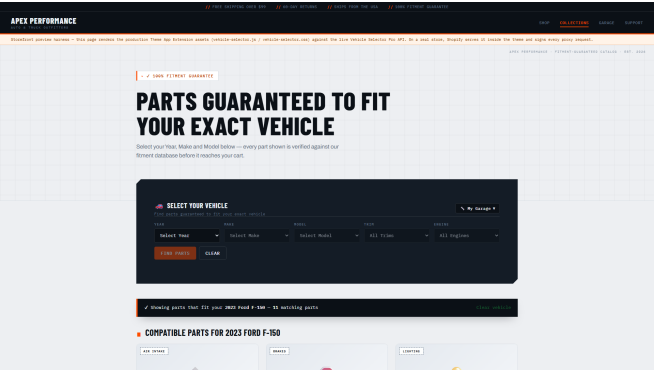


Zero ScriptTag API — pure Theme App Extension, signed App Proxy, and safe webhooks.

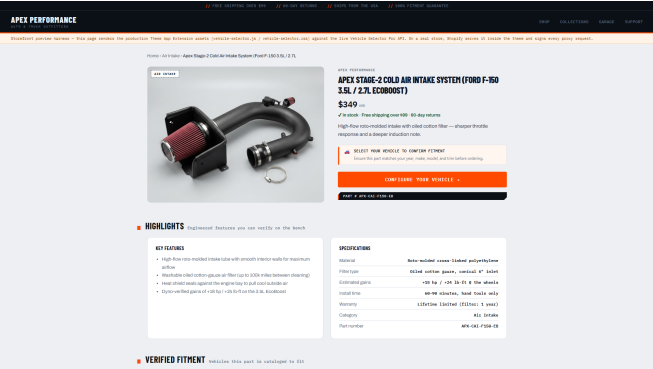
What shoppers see



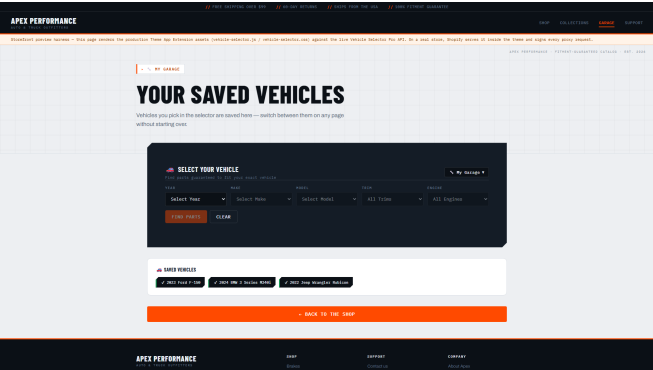
Cascading selector



Filtered collection

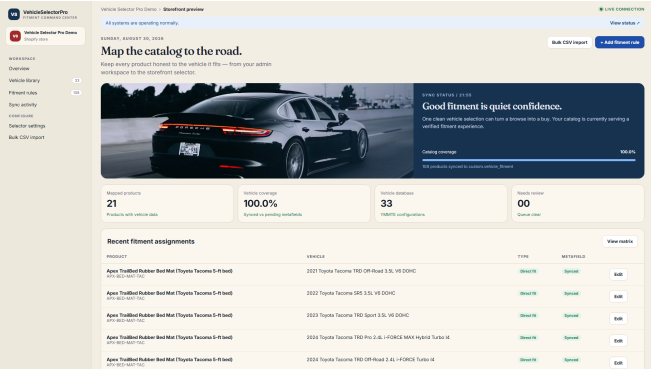


Fitment badge on PDP

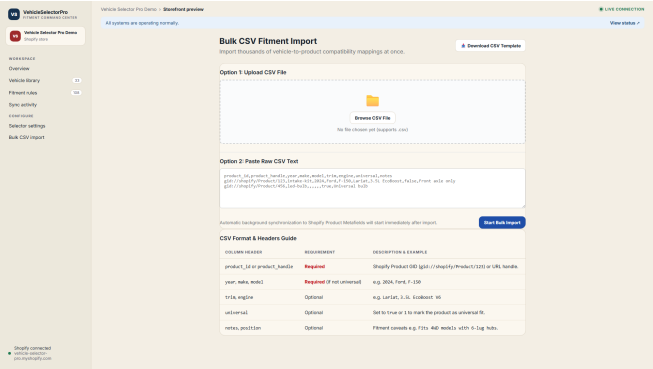


My Garage

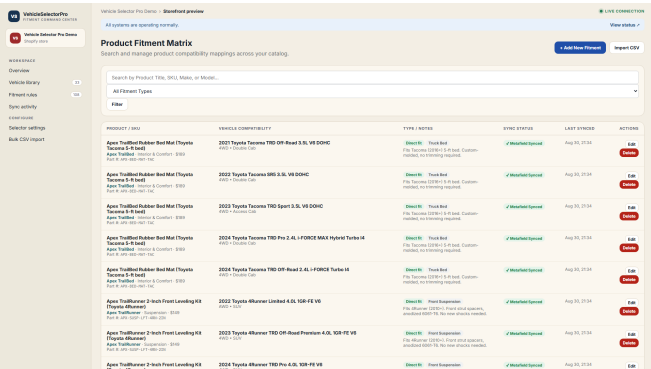
What merchants control



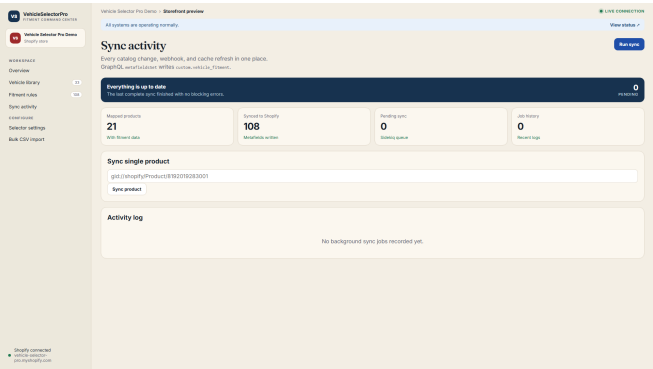
Command center



Bulk import



Fitment matrix



Sync monitor

Safe, multi-tenant, App-Store ready

- Standard OAuth 2.0 with encrypted tokens and per-shop data isolation.
- HMAC-SHA256 verification on every App Proxy and webhook request (constant-time).
- rack-attack throttling and CSP frame-ancestors locked to myshopify.com.
- Full GDPR coverage: data-request, redact, and shop/redact handlers plus a public privacy policy.

Engineering-grade stack

• PostgreSQL cache · SQLite dev

• Polaris · ERB

Deploy & install on your own store

- Create the app in Shopify Partners — copy the API key & secret, set the app and redirect URLs.
- Deploy the Rails app to Fly.io / Render with PostgreSQL, Redis and Sidekiq; boot with rails db:prepare.
- Register the App Proxy subpath /apps/vehicle-selector, then install on the store via OAuth (webhooks auto-register).
- Create the one-time custom.vehicle_fitment metafield definition in Admin (Product, JSON, Public read).
- Drop in the Theme App Extension blocks (vehicle selector + fitment badge) and publish the theme.
- Bulk-import fitments, run the sync, verify PDP badges & filters, and go live.

ROADMAP

A foundation built to grow

- Shopify Billing API for recurring and one-time plans.
- Supersession & OE-number matching for a richer part engine.
- Fitment confidence scoring and smarter universal/exact priority.
- White-label theming, translations, and merchant analytics.

Built at the ceiling of a 2018 ultrabook

- The entire project — Rails app, Shopify integration, storefront, diagrams, this deck, and both demo videos — was produced on one machine.
- ASUS ZenBook UX433FA · Intel Core i7-8565U @ 1.80 GHz · 16 GB RAM.
- Intel UHD 620 integrated graphics only — no GPU, no cluster; every frame of both demo videos was rendered on the CPU.
- The output was capped by tooling, not ambition. Given a real workstation, a GPU, and paid frontier models, the same foundation leaps a level.

Free tools, minimal hardware — maximum headroom

No paid AI models, APIs, or commercial tooling were used anywhere — and the whole result was built and rendered on a 15-watt 2018 ultrabook with no GPU. Both are deliberate. They prove what free tooling on minimal local compute already achieves, and set up a fast leap forward: with proper hardware and paid frontier models, the ceiling lifts dramatically.

READY WHEN YOU ARE

Let's take the next step together

- › **Try the live demo** — <https://vehicle-selector-pro.fly.dev/demo>
- › **Explore the merchant admin** — <https://vehicle-selector-pro.fly.dev/demo/admin>
- › **Open the repository** — <https://github.com/ai-dev-2024/vehicle-selector-pro>

Both the live storefront and merchant admin are running right now.